Day	Duration	Module	Topics Covered
Day 1	2 hrs	**1.Python Fundamentals**	**Introduction to Python:** Installing Python & Environment Setup (Anaconda, VS Code, Jupyter), Python Syntax & Execution Flow, Variables & Naming Conventions, Data Types (int, float, string, boolean), Type Casting & Input/Output **Control Flow:** Conditional Statements (if, elif, else), Operators, Loops (for, while), Loop Control (break, continue, pass), Iterators & Comprehensions **Data Structures**: Lists & List Operations, Tuples, Dictionaries, Sets, Practical Data Handling Exercises **Modular Programming**: Writing Functions, Arguments & Return Values, Lambda Functions, Modules & Imports, Creating Python Packages **Object-Oriented Programming (OOP)**: Classes & Objects, Constructors (__init__), Methods & Attributes, Inheritance, Encapsulation & Polymorphism **working with modules:** Type hinting (typing module), Pydantic models (BaseModel, validation)
Day 2	2 hrs		
Day 3	2 hrs		
Day 4	2 hrs	**2. LLM Landscape & Architecture**	Transformer architecture overview, attention mechanism (self-attention), embeddings, tokens and tokenization process Overview of frontier models (GPT-5 reasoning, Gemini, Claude 4, Grok), capabilities comparison, use-case mapping Open-source models, local vs hosted deployment, inference basics, model selection criteria
Day 5	2 hrs		Economics: token-based pricing, latency vs cost trade-offs, API vs self-hosting, GPU/infra considerations, scaling strategies Context window limits, hallucination causes, mitigation strategies, Fine-Tuning & Instruction Tuning,
Day 6	2 hrs	**3. Prompt Engineering**	Prompt design principles, instruction clarity, zero-shot and few-shot prompting, role-based prompting
Day 7	2 hrs		Advanced techniques: chain-of-thought, self-consistency, prompt chaining, programmatic prompting pipelines
Day 8	2 hrs		Structured outputs, JSON schema enforcement, function calling concepts, Pydantic integration for validation
Day 9	2 hrs		Adversarial prompting, step back prompting, prompt injection techniques, self-prompting

Day 10	2 hrs	**4. Knowledge Augmentation**	RAG fundamentals, embeddings for retrieval, vector databases (FAISS/Chroma), indexing pipelines
Day 11	2 hrs		Chunking strategies (fixed, semantic, recursive), metadata handling,
Day 12	2 hrs		RAG evaluation: RAGAS framework, metrics (faithfulness, relevance), error analysis, improving retrieval quality
Day 13	2 hrs	**5. AI Agents and Agentic AI: LangChain, LangGraph, CrewAI**	What are AI Agents and Agentic AI ? , needs, core properties including autonomy, tool usage, decision-making, and iterative reasoning.
Day 14	2 hrs		Introduction to ReAct paradigm, Tool-Using Agents & Memory-Driven Systems, Function calling / tool invocation, Memory: Short-term (conversation state) , Long-term (persistent memory)
Day 15	4 hrs		AI workflow frameworks, LangChain ecosystem basics: chains, prompts, tools, memory.
Day 16			
Day 17	4 hrs		LangGraph: stateful workflows, DAG-based agent design, multi-step reasoning flows, HIML, Introduction to MCP (Model Context Protocol): architecture, connecting tools and agents
Day 18			
Day 19	2 hrs		Multi-Agent Collaboration and CrewAI Framework, Multi-agent orchestration and communication, CrewAI : team-based AI collaboration, Goal decomposition, task scheduling, and monitoring, Human-in-the-loop supervision for decision approval Introduction to deployment concepts for agent-based systems and best practices.
Day 20	2 hrs		
Total Hours: 40 (total 20 working days)			